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Article

Empowering Residents: A Theoretical Framework for Negotiating Surveillance Technologies

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Abstract

This article aims to develop an understanding of negotiation as a proactive user behaviour in response to persuasive, surveillance-enabling technologies. We work with a case where the ambition is to use maps to persuade residents in marginalized residential areas to take co-ownership of the future of their neighbourhood. However, we argue that by negotiating persuasive technologies, residents can domesticate and "reconfigure" these, thus we need to broaden the dominant understanding of persuasion (Fogg 2003) to include the process of negotiation. Inspired by Actor-Network Theory, we argue that we need a framework that takes into account negotiations of other actors than the persuasive designer to bring a more nuanced and active understanding of responses to surveillance technologies. This article suggests that negotiation can be both user empowering as well as a productive measure in an ongoing designer-user relation. While the designer persuades, the user negotiates, and this dynamic relation contributes to potentially better technology development.

1. Introduction

The aim of this article is to develop an understanding of negotiation as a proactive user behaviour in response to persuasive, surveillance-enabling technologies. Often, surveillance technologies and practices are seen as being undesirable, antithetical to democracy and individual autonomy (Monahan 2010) and, as a response to this, some theorists have argued that we should evaluate the success or failure of new technological systems with regards to the level of citizen participation and democratic potential (e.g. Winner 1977; 1986; Martin 1999). These potentially totalitarian aspects are most often associated with surveillance as a large-scale structure or system that dominate individuals and groups. This unfavourable power relation makes participation difficult, if not impossible, and this makes surveillance seem incompatible with democracy. Our approach is to explore a small-scale design process and we emphasise the empowering aspects of surveillance to support user involvement.

We work with a case where the goal is to use maps to support residents in marginalised residential areas in taking co-ownership of their neighbourhood's future. Moreover, the aim is to persuade them to express this ownership using surveillance technologies that make it possible to connect certain experiences to certain places and which can be used in strategic planning for the marginalised residential areas. Here, we argue that, by negotiating technologies, residents can domesticate and "reconfigure" these; subsequently we need to broaden the dominant understanding of persuasion using technologies (Fogg 2003) to include the process of negotiation. Inspired by Actor-Network Theory, we argue that we need a framework that

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takes into account the negotiations of actors other than the persuasive designer to bring a more nuanced and active understanding of responses to surveillance technologies. This article suggests that negotiation can be both user empowering as well as a productive measure in an ongoing designer-user relation, where designers as well as users negotiate the technologies. This dynamic relation, we argue, contributes to potentially better technology development.

The focus of the article is surveillance-enabling technologies in "mixed spaces", i.e. spaces that combine the online and offline worlds. Just a few years ago, advanced monitoring and tracking of people, both in virtual and physical space, was difficult and required special equipment. Today, pervasive surveillance technologies are a part of everyday life for many people, especially in the city and on the web. In cities we meet a wide range of mobile devices, e.g. smartphones and GPS navigation units, which distribute surveillance over a number of locational technologies and make it possible to track people outside as well as inside. Similarly, there are many ways to track online activities, as digital "footprints" are in many cases persistent. Moreover, social networking sites and services often facilitate communities that are based on the premise of sharing personal information, including activities, preferences and current whereabouts. Consequently, offline and online activities frequently mix, as web applications and services are integrated with offline practices. Examples of such mixed spaces are mash-ups that involve Google Maps. Here, cartographic data provided by this web application is combined with the data from different kinds of websites.

With surveillance technologies and practices playing such a central role for the activities in these mixed spaces it is very interesting to dwell on the concept of surveillance. Is this a frightening development? Does the advanced monitoring and tracking technologies invade our privacy? Are people lured into unfavourable power relations? On the other hand, do the potential gains outweigh the eventual losses? Are surveillance technologies helping us, protecting us, empowering us and offering new possibilities? As indicated here, the way we theoretically *understand* surveillance is very important for the given frame of answers to these questions. In the second section of this article, we discuss this conceptual issue and introduce the concept of "participatory surveillance".

The third section is concerned with the potential struggle between persuasion and negotiation. It is well known that surveillance technologies have been thought to structure people's behaviour. Classic examples are buildings inspired by Bentham's infamous Panopticon design, which was intended to discipline its inmates, patients, students, or workers (Bentham & Bozovic 1995; Foucault 1975). In a similar way, today, CCTV and other technologies are among other purposes intended to regulate and structure the behaviour of citizens, customers or employees (Norris & Armstrong 1999) and can thus be considered a persuasive technology similar to mobile, locational technologies designed to encourage certain behaviours (Fogg 2003). However, users also negotiate these (Ball & Wilson 2000) and, in the case of surveillance, reverse the power relation by transforming devices into technologies of user empowerment (Koskela 2004). Here, we introduce the concept "networked persuasion" which can help us identify different agencies at play in the persuasive design process and, in turn, help us develop design methods that take into account negotiations of technologies. In this section we also introduce our case, a current research project at Aarhus University, which is about creating maps using surveillance technologies. These maps connect experiences and behaviour to places, and the project's aim is to develop a way to use these in the planning of initiatives in marginalised residential areas.

The fourth section takes the concepts developed in the two previous sections and discusses them in the context of our case. Using the concept of networked persuasion, we argue that, to avoid manipulation in the mapping process, it is necessary to carefully identify and evaluate different stakeholders' approaches to the mappings as well as the residents' negotiations of the mapping technologies, and from that trace a variety of other actors and agencies at play. In order to study different approaches and negotiations, we need to develop a mapping method that makes visible different conditions in the persuasive process, and

here we suggest a set-up where residents in marginalised areas are asked to carry GPS-enabled smartphones. This makes it possible for them to comment on their activities, and the result will be user-generated maps that visualise experiences and behaviour in the given area. We use the concept of participatory surveillance to emphasise and design for the empowering and social aspects of these mappings. We argue that persuasive, surveillance-enabling mappings can be potentially empowering for residents if they also enable them to communicate their negotiations of the process as part of creating and sharing identities. Accordingly, creating content for maps over marginalised residential areas can work as an act of reclaiming the neighbourhood. To support such empowering potentials we suggest workshops as a way to invite residents to discuss and re-design the maps, so as to more adequately visualise their experiences and activities as these change over time.

2. Surveillance and Empowerment

When we study and discuss theories of surveillance, a reasonable way to begin is by addressing the problem of definition. The surveillance literature offers many definitions, often in a form where authors state how the concept is used and should be understood in a given context (e.g. Rule 1973; Marx 1988; Lyon 2007). This is, of course, very useful in the given context, but it leaves open more fundamental questions concerning how to define concepts and, in particular, how to define surveillance. In this article, we do not intend to fully address this interesting but also extremely comprehensive issue. In the following, our ambition is to position the concept of participatory surveillance among other approaches in the surveillance literature.

2.1. A Pluralistic Approach

The Aristotelian way of defining involves the disclosure of necessary and sufficient conditions of the concept to arrive at an unambiguous, essential definition. This classical theory has often been seen as the ideal way of defining, as it promises to arrive at the core content of concepts, i.e. explaining what something "really means". It is an essentialist approach to definitions that has been disputed by modern philosophical theories[1] and likewise does not seem to be practiced in the field of surveillance studies. Rather, the surveillance literature consists of many types of perspectives on monitoring practices and technologies, representing a wide variety of academic disciplines as points of departure. Our ambition here is to pursue this perspectivism further, first by elaborating on different perspectives within surveillance studies and, second, by introducing the idea of participatory surveillance. Consequently, we argue for a pluralistic approach to surveillance that focuses on relations and contexts.

Even though the two most dominating perspectives stem from fiction (Big Brother) and philosophy (Panopticon), the most notable academic discipline relating to surveillance has traditionally been sociology. However, due to the pervasiveness of surveillance and privacy-related issues, scholars from many other academic disciplines ranging from law and the humanities to the social sciences have contributed to the literature and, today, surveillance studies is an interdisciplinary field of research. Furthermore, non-profit organisations, e.g. *Privacy International*, anti-surveillance activists and (performing) artists, such as the *Surveillance Camera Players*, take part in the study and discussion of surveillance. This pluralism indicates that surveillance technologies and practices have been studied, discussed and explained from many points of departure. Also, when we take a look at the contexts in which surveillance plays a role, a manifold of different relations, ways and types of situations appear. These contexts offer different perspectives on surveillance that involve a wide variety of conceptions and discourses that are more or less detached from Big Brother and the Panopticon.

[1] The classical theory has been disputed within contemporary philosophy, and it has been challenged by a variety of alternative theories, e.g. the prototype theory. Consequently, the discipline of defining is part of an ongoing debate within philosophy, but it is not possible to further address this issue here.

2.2. Perspectives on Surveillance

To illustrate these many possible contexts and to give depth to the concept of surveillance, we have developed elsewhere a list of perspectives which represent different viewpoints and vocabulary (Albrechtslund 2008). In what follows, we will summarise these many perspectives by grouping them according to three overarching categories. Based on the surveillance perspectives that have been described to date, we propose the following categories: 1) study perspectives, 2) use perspectives, and 3) cultural, social and existential perspectives. These categories of perspectives describe how surveillance is understood as a concept, either tacitly or explicitly, and how this understanding frames the discussions, arguments, opinions and uses of surveillance. Thus, this is not meant to be an exhaustive list, but a tool to emphasise the manifold ways of thinking about surveillance.

The first group of perspectives includes the many ways that surveillance has been interpreted in the research literature. Obviously, Foucault's reintroduction of the Panopticon as illustration of discipline in the modern society has played a central role for the understanding of surveillance as a power mechanism in society. Moreover, the idea of surveillance as involving power relations – for control or care – has been further developed by sociologist such as Gary T. Marx, David Lyon and others in the sociological tradition of surveillance studies. Other important research perspectives include legal issues, often relating to privacy, and ethical issues concerning the change in relations between people caused by surveillance with regards to power and knowledge, which results in a changed space for ethical actions.

The second group of perspectives can be interpreted as different kinds of practical contexts or "professional" approaches to surveillance. Rather than theorizing about surveillance, these perspectives have to do with different kinds of concrete monitoring practices. The professional perspectives include practices relating to security, safety and prevention, which can be oriented toward private, corporate and governmental institutions. Often, these perspectives derive from practical situations where surveillance technologies, e.g. closed-circuit television, are deployed to protect people, things or interests. Thus, this way of understanding surveillance implies that monitoring practices are conceived of as an instrument for certain purposes, e.g. security, safety and prevention of crime. Besides being a passive means of protection, professional perspectives on surveillance can be active, e.g. in the form of spying or investigations. Here, information is pursued using surveillance technologies and practices.

The third group of perspectives relates to cultural, social and existential issues. Some of these perspectives have been very influential on our everyday understanding of surveillance as well as for the basic arguments in the public debate. It almost goes without saying that Big Brother has been the most important and defining metaphor for our understanding of surveillance, sometimes accompanied by borderline paranoid conspiracy speculations. Besides this infamous character invented by George Orwell, a number of other less noticeable but very interesting perspectives must be mentioned. These include playing, gaming and leisure activities, which can be based on surveillance technologies and practices. A simple example of such an activity based on a type of surveillance practice is the children's game "hide and seek", while more technologically based examples could be *Monopoly Live*[2] and *Can You See Me Now?*[3] Furthermore, surveillance facilitates the enforcement of rules in sports, as athletes must necessarily be monitored closely to measure and compare their performances. From a cultural point of view, surveillance is an artistic way to deal with issues of society, modernity, transparency, etc. in works of art

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2 Hasbro (the game publisher) introduced *Monopoly Live* as part of a promotional drive in 2005: "We have turned London into a real-life playing board and real taxi cabs into real-life playing pieces" (see http://www.monopolylive.com/). A number of taxis were equipped with surveillance technology to track their whereabouts. Based on the recorded traffic, the value of apartments and hotels went up or down.

3 This game which takes place simultaneously online and in the streets was created in collaboration between Blast Theory, an artists' group, and Mixed Reality Lab, University of Nottingham, England. More information can be found on the project website: http://www.blasttheory.co.uk/bt/work_cysmn.html.

such as installations (e.g. *Nanobots*[4]) and happenings (e.g. the *Surveillance Camera Players*[5]). Surveillance is also a theme of suspense and fascination in literature, poetics, computer games, cinema, etc. as well as an issue in film theory involving the audience as watchers/voyeurs and the movie as spectacle. This exhibitionist-voyeuristic perspective extends to an understanding of surveillance as a way to display oneself for and/or (secretly) watch other people for (erotic) pleasure. Finally, surveillance can be interpreted as an integral part of human existence, both as watching and being watched, as we become aware of our own self by seeing others watch us. Moreover, by watching other people, we build our own identity; thus, surveillance is a key existential concept[6] in understanding human life.

As is apparent from these numerous perspectives, the way in which surveillance is conceived and the way to describe it differs according to the context. For example, negatively oriented perspectives such as the Big Brother and paranoid perspectives are often associated with a corresponding negative vocabulary, e.g. "hidden", "slippery slope", "intrusion", and so forth. Similarly, positive and more or less "neutral" perspectives describe surveillance technologies and practices in terms corresponding to how the situation is conceived. Consequently, it is difficult to generalise or find a "God's-eye" perspective on surveillance which encompasses all the contexts mentioned in the above. Rather, the meaning of surveillance seems to be dependent on the eyes of the beholder (Albrechtslund 2008) and subsequently, as opposed to an essentialist definition, we suggest a pluralistic approach to the question of defining surveillance which allow for the manifold perspectives we find both in the scholarly and everyday use and understanding of the concept.

2.3. Participatory Surveillance

With this pluralism and perspectivism in mind, we propose the concept of participatory surveillance[7] to develop the empowering aspects of monitoring technologies and practices. In particular, our focus is on power relations and the social role of surveillance. In connection with power relations, we elaborate on the idea of empowerment of the watched and the controlled visibility as a way to build identity and also as a way to negotiate surveillance technologies. Our suggestion is that surveillance is not only an unfortunate "side effect", as it is sometimes claimed (e.g. Gross & Acquisti 2005); it is also an integral and productive part of social life in the mixed space of the web (Albrechtslund 2008) and the city.

The Panopticon and Orwell's *Nineteen Eighty-four* are models that represent a hierarchical conception of surveillance. Such a conception reduces the person under surveillance to a powerless, passive subject under the control of the gaze. The idea of participatory surveillance is to introduce a very different, almost opposite, conception. Here, the power relation can be beneficiary for the person under surveillance, as the visibility facilitated by monitoring technologies and practices offers opportunities for empowerment. We

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4 *Nanobots* is an interactive installation by the German artists' group *Beobachter der Bediener von Maschinen* (BBM) which is intended to involve the visitors in an experience of being pursued and monitored by robots (see http://www.bbm.de/projekte/nanobots.html).

5 The *Surveillance Camera Players* is a group based in New York City which uses surveillance cameras and theatrical effects to protest surveillance (see http://www.notbored.org/the-scp.html).

6 For example Jean-Paul Sartre's discussion of the constitution of subjectivity in *L'Être et le néant* (1943), illustrated by a story about a voyeur peeping through a keyhole. This act of surveillance puts the watcher in a pre-reflective state of mind, because the person's attention is directed towards the situation on the other side of the door. Only when steps are heard in the corridor, signifying that another person is approaching, the person looking through the keyhole becomes self-aware as the object for someone else's gaze and, thus, enter into a reflective state of subjectivity. As this example shows, surveillance can play an important role for the existential constitution of subjectivity, and an elaborated existentialism could thus complement a sociological understanding of subjectivity.

7 The concept "participatory surveillance" has been used before by Mark Poster (1990) and T.L. Taylor (2006). Poster argues that today's circuits of communication and databases constitute a superpanopticon, where individuals are not just disciplined but take an ever more active part in their own surveillance by continuously contributing information to databases. Taylor uses the concept to study collaborative play in the online computer game *World of Warcraft*, arguing that norms and coercion, play and pleasure are not necessarily antithetical.

have elsewhere studied online social networking and concluded that practices found here reveal surveillance as a potentially empowering and integral part of social life (2008). Rather than being a system of repression, surveillance can be a way to build identity and make sense in the lifeworld for those being watched, since monitoring and registration also facilitate new ways of socializing with friends and working with colleagues. This changes the perceived role of the watched from passive to active, since surveillance in this context offers opportunities to take action, seek information and communicate. Online social networking therefore shows that surveillance – as a mutual, empowering and subjectivity building practice – can be fundamentally social.

To illustrate this perspective on power relations, we point to Hille Koskela's discussion of the use of webcams, TV shows and mobile phones (2004). She introduces the concept "empowering exhibitionism" to describe the practice of revealing one's (very) personal life. By exhibiting their activities, people claim "copyright" to their own lives, as they engage in the self–construction of identity. This reverts the vertical, Big Brother-inspired power relation, as visibility becomes a tool of power that can be used to rebel against the shame associated with not being private about certain things. Thus, Koskela argue that exhibitionism is liberating, because it represents a refusal to be humble.

An important aspect of this perspective on surveillance is the idea of participation. To participate means to engage in something, but it is not necessarily something we do out of individual desire or pleasure. Examples could be found in all contexts of life, including work–related situations and charity efforts. However, participation as an engaging act is voluntary and must be well–defined in relation to the pseudo–participation we know from the Panopticon and Orwell's *Nineteen Eighty–Four*. The Panopticon is set up in a way for the prisoners to take part in their own surveillance by internalizing the gaze of the watcher, and in Orwell's novel the citizens of Oceania ends up taking part in their own (and others') surveillance in their "love" of Big Brother. Here, the self–surveillance is inflicted on the people watched, as they are caught up in a power relation (Foucault 1975) or as a result of the brainwashing carried out by the Ministry of Love (Orwell 1949). Both of these disciplinary practices are disempowering and, thus, disengage the subject of surveillance. Therefore, concepts such as participatory Panopticon (Cascio 2006) are contradictory or, at best, redundant, if the internalizing of the gaze is interpreted as a form of pseudo–participation.

The concept of participatory surveillance is closely associated with the idea of sharing. It is important to not automatically assume that personal information data submitted in surveillance relations are only commodities for trading, because implicit in this understanding is that being under surveillance is undesirable. When we look at online social networking and related practices, participating is also about the act of sharing yourself – or your constructed identity – with others. Accordingly, the role of sharing should not be underestimated, as the personal information people share – profiles, activities, beliefs, whereabouts, status, preferences, etc. – represent a level of communication that neither has to be told, nor has to be asked for. It is just "out there", untold and unasked, but something that is part of the socializing in mediated publics. One of the findings in a Pew Internet & American Life Project report (Lenhart & Madden 2007) is that a great majority of teens use online social networking to keep in touch with friends they rarely see in real life. In this case, participatory surveillance is a way of maintaining friendships by checking up on information other people share. Consequently, the concept of participatory surveillance stresses that the surveilled *acts* with the technology. In the next chapter we further elaborate how this acting can be understood as a negotiation of persuasive technologies.

3. Mapping as Persuasive Intervention

In this section, we first present the idea behind Persuasive Technologies (PT) and discuss some limitations in relation to the development and design of these. Next, we introduce a theoretical framework, which we call "networked persuasion", and here we seek to combine PT with elements of Actor-Network Theory

(ANT). We then use networked persuasion to elaborate on the power struggle between the designers' persuasive approaches and user practices, and thus to overcome the potential struggle between persuasion and negotiation. Lastly, we present our case in which maps are used for persuasive interventions in marginalised residential areas.

3.1. Persuasive Technology

A fundamental characteristic of PT is the aim of intervening with an intention of changing people's attitudes and behaviour (Fogg 2003). According to Fogg, technologies can be designed with specific purposes in mind, which can then be employed to persuade users. Fogg distinguishes between "macrosuasion" and "microsuasion", which represent two different levels of persuasion. The first level addresses technologies that are fully designed for persuasive purposes, e.g. mobile technologies to boost physical activity or manage a chronic disease (Boland 2007; Damen 2007). The latter are technologies that serve some other purpose or function but still have an element of persuasion, for example running watches that are made to self-monitor the runners' own performance, including heart-rate, speed and distance. Moreover, it motivates runners to compare results and thus socialise around the running activity.

The idea of persuasive technologies – whether macro- or microsuasion – presupposes that the use of technologies to some extent must correspond to or reflect the designer's intentions. Furthermore, this correspondence between design and use must be more or less controlled. If, for example, a mobile technology that seeks to persuade users to exercise more is in fact used to socialise, but without changing moving patterns, then the designer's persuasive intentions do not correspond to the use context. This example and other similar situations can be a problem for the theoretical idea of designing technologies, in so far as the ambition is (also) to persuade the user.

To some extent, the idea of persuasive technology design shares this problem with technology design in general. Naturally, if technologies are designed to serve certain functions (communication, information, entertainment and so forth) then the success or failure of the design can depend on how well the actions that the design is built to promote are actually carried out in the use situation. However, persuasive technologies go beyond merely designing for certain actions, as the designer intentions are more ambitious. Persuasive technology design aims at influencing attitudes and, as such, the central idea is associated with theories of value-conscious design. Such theories, for example Value Sensitive Design (Friedman 2004; Friedman, Kahn, and Borning 2002), aim at "building in" values in technologies. We have elsewhere discussed that such ambitions are faced with a "positivist problem" (Albrechtslund 2007).

We argue that a similar critique regarding theories of persuasive technology design is appropriate. The core of the problem is the relation between designer intentions and the eventual use of the technology. It is positivist, because the default assumption is that the intention can be built into the technology and this will – more or less – correspond with the use of it. We consider this to be a problem, because design and use do *not* seamlessly correspond. Several other actors are constantly negotiating the persuasive design and, consequently, if we want to design for persuasion, we must take into account these actors in our framework. We propose that this can be done by connecting to ANT, since it is a method that enables us to see the design situation as a network of actors all partly constructing the network (Glud et al. 2009). Leaning on ANT will thus help us obtain a more nuanced understanding of the situation in which we wish to make a persuasive intervention.

3.2. Networked Persuasion

ANT seeks to trace "matters of concern", which are uncertain, negotiated, and objective agencies connected in gatherings or "assemblings" (Latour 2005: 114). Tracing matters of concern, ANT offers a different view on actors, groups and action, and reach from that a different understanding of what it means to make someone do something – or to persuade. In ANT an actor is defined as a mediator and this means

that an actor is any*thing* that transforms a state of affairs by making a difference. Consequently, actors include humans as well as non-humans such as animals, buildings, technologies, facts, emotions etc.

Since ANT is concerned with tracing agencies, it is important to be able to distinguish between acting and non-acting entities (mediators and intermediators) (Latour 2005: 39). Here, an intermediary is defined as an entity that transports a meaning or element and a mediator is defined as an entity that translates it or *negotiates* it. In the first case we can deduct the output from the input, however, in the second this is not possible. Translation is here understood as: "a relation that does not transport causality but induces two mediators into coexistence" (Latour 2005: 108). This results in an understanding where an actor is always part of a composition of several agencies and is therefore never alone when acting. Consequently, there is a dislocation in making someone do something, except by creating a restraining and stable network through several translations.

To be able to trace a diversity of actors, ANT focuses on studying controversies (Latour 2005: 23) or, in another word, negotiations. This means that we need to take several conflicting agencies into account. Here, it is more important to focus on how these agencies act, as mediators or intermediaries, rather than to identify which agency to choose or if these agencies have the figuration of groups, individuals or concepts. Consequently, to study controversies is to study negotiations of what agencies are at play, and here, surveillance technologies can be used as a way to stir the pot and facilitate such negotiations (Glud et al. 2009: 7).

Consequently, connecting to ANT entails that it is necessary to abandon any ambitions of *a priori* predictions of how technologies will be used and instead focus on concrete use situations and networks of actors where negotiations take place. Thus, a theory about designing persuasive technologies can with advantage be developed to focus on technologies as part of networks of actors. In this article, we suggest ways of thinking about surveillance technologies in combination with an ANT-inspired theory of persuasive design. We use the concept "networked persuasion" to describe a mapping process which involves all relevant stakeholders and, hopefully, can empower residents.

3.3. Persuasion and Negotiation

We have argued that in order to plan for persuasive interventions, we need to focus on negotiations to be able to trace a variety of agencies. Keeping with the theme of surveillance, this could be illustrated with a mapping example that builds on the case that we will elaborate later. Here, surveillance technologies are used to map the experience and behaviour of residents in marginalised urban areas. In this case the design situation cannot adequately be described in a single purpose. Several actors are involved, including the landlord, the municipality, the shopkeepers in the area, the politicians and police, and these actors might have very different perspectives about the mapping with regards to expectations, opinions and reasons for wanting to employ surveillance. Thus, several types of persuasive motivations and approaches related to the technologies can be identified, including those that seek to influence current and future residents' perception of security and safety, and to avoid criminal behaviour.

Mapping experience and behaviour in urban areas can also be interpreted as a way to exercise power. Since actors constantly negotiate each other's actions, such actions can be met with counter power. In the case of CCTV, the negotiation manifests itself when people try to avoid the surveillance, e.g. by covering their faces or staying out of the cameras' reach. A violent version of this negotiation can be seen when people purposely try to make the cameras dysfunctional or even damage them. More creative examples of negotiations are the many artistic ways of addressing issues relating to GPS technology. Thomas Berenzinski (Wollan 2009) and Jeremy Wood (2009) represent examples of people who use the GPS in their smartphone to create drawings of, for example, lions and elephants on maps. Others use the technology to write messages, for example, messages in relation to getting sponsorship for a marathon (Rice 2009).

When people negotiate, technologies will sometimes be used in new and perhaps even unexpected ways, which can be described as a reconfiguration of the technologies. Consequently, persuasive technologies are redefined according to the use context, which can be interpreted as a practical "deprogramming". For example, a speed bump is designed to persuade drivers to slow down in certain areas, but if these bumps instead are used at night by young drivers to make spectacular – and very dangerous – car stunts, then the technology is reconfigured and the persuasive design is challenged. This is, of course, a rather unfortunate negotiation and our ambition in this article is to point out more beneficiary ways of reconfiguring technologies. We suggest that negotiation in the form of reconfigurations can enter into a dynamic relation between user and designer, which has the potential to inform the design process. Later we argue that inviting, supporting and including negotiations in the design process creates a potential for empowering the groups being surveilled.

3.4. Mapping as Monitoring Strategy in Marginalised Residential Areas

The aim of our project is to use maps as platforms for persuasive intervention in marginalised residential areas in Denmark. These areas were built with certain social ideas in the 1960s early welfare society. However, during the last 50 years, tensions have arisen due to social problems, resulting in vandalism and crime. The area in which the mappings will take place is Gellerupparken in Aarhus. The housing blocks in this area are made of concrete and were built in the period 1968-1972. The blocks consist of 1,776 apartments and were home to 5,863 residents in 2008. 56% of the residents in 2007 were not part of the labour market and 88% of the residents in 2008 were immigrants or second-generation immigrants (Programbestyrelsen for dialog og balance i udsatte boligområder 2008). In the project we use surveillance technologies (smartphones and mapping software) to create maps that connect experiences and behaviour to certain places in the areas described above. The idea is that such maps can contribute to the planning of initiatives with the intention of motivating or persuading the residents in taking better care of the built environment by allowing them to communicate co-ownership over their local area's future. Our central questions are: How is mapping as a monitoring method negotiated in practice? How can these negotiations be made visible in maps? How can residents be engaged in the production of the maps?

Maps can be described as visual representations emphasizing relations between spatial elements. This is done in different scales and orientations, focusing on different attributes, which makes mapping a process of selecting what information fits the aim of the map (Gronim 2001; Monmonier 1996). In other words, maps do not simply show the world "as it is", but rather constitute a means to describe something in a certain way for others. As a result, they have far reaching consequences for the institutions they are part of and the individuals they are connected to (Curry 1998). An example is Henricus Martellus' world map (1490) which seemed to prove that the distance between Europe and China was not that long which, in turn, offered Columbus the opportunity to persuade Ferdinand of Aragon and Isabella of Castile to support a travel to China (Barber 2010).

However, using maps as persuasion is not a just a medieval phenomenon. Today, Geographic Information Systems (GIS) and Geographic Positioning Systems (GPS) are used to design maps that visualise people's patterns of movement. Such maps are often created to fit the aims of a certain project, whether this is to develop an urban design strategy with a goal of, for example, making young people use the sports areas in the city more, or to convince stakeholders of projects that a re-design of a specific residential area is needed to attract new residents. Consequently, mapping can represent the world in a way that fits the purpose of the action one wishes to carry out or want others to carry out. This means that maps can be used not only to influence how people understand but also how they are involved in a certain place. This understanding shows that mapping technology demonstrates a partiality as well as a possibility for critical usage (Robbins 2003).

We have argued that the negotiation is important if we want to make persuasive interventions. As such, we believe it is important to investigate how the mapping process is negotiated in practice and how this

negotiation can become part of the maps. To embrace negotiations, Renée E. *Sieber* argues for a rewiring for GIS/2 that integrates local knowledge and preserves disagreement rather than reducing it. The application of methods that embrace negotiations have already been done by the artist and designer Christian Nold, who involved citizens in creating emotion maps of their neighbourhood (2009a, 2009b), and Rattray, who has explored how GIS can be used to investigate the effectiveness of universal design and accessible space (2007). Moreover, initiatives such as OpenStreetMap that allow everyone to view, edit and use geographical data together also make maps into accessible platforms for negotiations.

In relation to the planning of interventions in marginalised residential areas, the Danish government announced a strategy against ghettoisation in Denmark, which included seven focus areas, three of which are especially relevant to our project: 1) safety and neighbourliness, 2) efficient organisation and 3) communication (Programbestyrelsen for dialog og balance i udsatte boligområder 2008). The year after this, the municipality of Copenhagen created a "Safetymap", which shows citizens' experience of safety in different areas in Copenhagen (Davidsen-Nielsen 2009). The map was designed using a "safety barometer" where citizens can indicate their level of safety in different areas. It was used to prioritise the police effort and did not include the citizens' diverse negotiations of the map and the built environment.

We argue that it is important to include the citizens' negotiations in order to involve them as visible active participants in the development of the city. In relation to this, the Council of Crime Prevention argued that we need to "...increase the individual's responsibility and psychological co-ownership to areas, furniture, buildings and means of transportation" (Det Kriminalpræventive Råd 2006). Consequently, it is not enough to organise the effort to intervene; it is also important to support different types of ownership in relation to the built environment. Combined with the three points mentioned above, this means that there is a need to develop tools and methods that: 1) make it possible for citizens to communicate a diversity of experiences besides safety and neighbourliness in relation to different areas, and 2) are useful as platforms from which stakeholders in the areas can organise intervention. In relation to our case, the aim is not to create co-ownership, rather we wish to support what is already there. However, making available specific tools and methods that allow the residents to express their co-ownership might result in new ways of feeling ownership towards the neighbourhood. We argue below that the theoretical framework we have introduced is useful to develop a set-up for designing maps where the residents in marginalised residential areas are invited to act co-responsible for the creation of the image and future of their neighbourhood.

4. Mixing Spaces: Empowering Residents

In what follows, we discuss how to understand and develop GPS mappings in marginalised residential areas in the context of participatory surveillance and an ANT-inspired theory of persuasive design, which we call "networked persuasion". The power relations and issues relating to networked persuasion are first discussed in the context of our case and suggested as a theoretical framework. Participatory surveillance is then suggested as a way for understanding negotiation as a productive and empowering perspective on mappings in marginalised residential areas. Building on the developed framework, we then suggest a set-up for creating mappings in marginalised residential areas which have the potential to empower residents while working as persuasive platforms.

4.1. Mapping Persuasive Networks

The aim behind the project "Mapping as Monitoring Strategy in Marginalised Residential Areas" is to influence attitudes and change behaviours and, as such, the ambition is to design maps for persuasive intervention. The maps have two purposes: 1) to involve residents in the development of their neighbourhood, and 2) to use this involvement as part of strategic planning with a persuasive purpose. In relation to the discussions earlier in this article, maps as persuasive technologies pose at least two challenges. The first concerns the power relations between the involved actors, which include designers, the landlord, the municipality, the politicians, the police and the residents. Here, the maps will be the stage

of struggles between persuasive approaches and different kinds of responses where these are negotiated. The second challenge has to do with the ethical question concerning the act of persuasion; namely, what is the difference between persuasion and manipulation?

With regards to power relations in the mappings of marginalised residential areas, there are several actors with different types of persuasive approaches who are all part of the same network. Here, it is appropriate to subdivide some of the identified actors and work with a more nuanced understanding of these. Who is acting? What actions are they doing? Who or what affects them to act as they do? The different stakeholders such as the researchers/designers, the politicians, the police, the municipality and the landlord might all have different reasons for mapping a marginalised residential area and, therefore, they might act in different and even conflicting ways. The researchers/designers' agenda might be to develop theory or perhaps produce publishable research papers. The police and municipality might base their participation on an interest in improving their own working conditions and environment. The landlord can have an interest in cutting expenses to sustain and renovate the urban area, and politicians might try to increase their own popularity among the voters. All these different motivations might result in different approaches to the marginalised urban area. Moreover, non-human actors also act, e.g. the buildings in the marginalised urban areas might look neglected and thus act in a way where they do not support the residents in taking care of them or, perhaps even worse, encourage residents to vandalise them.

As this example shows, mappings of marginalised urban areas represent a number of diverse motivations and approaches to persuasive design. These motivations and approaches indicate that persuasion, as communication and power exercise, is the result of diverse actors and agencies at play – some of which are non-human. In a similar way, the residents being mapped have different motivations and approaches. Among these can be attempts to ignore the persuasive technologies or outbursts of destructive behaviour either directed towards the mobile device, the urban environment or other people. These responses are ways of negotiating the task and as a result should not be ignored but taken into account. In a similar way the violence and crime in these areas can be seen as reactions caused by disempowerment imposed from external and/or comprehensible pressures. The responses can also be less violent. For example, the residents might be interested in marking or tagging places on the maps to claim ownership over or comment these in new ways (as in graffiti), or they might be interested in playing with the surveillance technologies. Since our aim is to involve the residents in the development of their neighbourhood, our focus is on mapping these actors' locations and experiences. However, using ANT means that by doing that we follow the residents' accounts of what actors are at play in the local area. As a result the maps will display a diversity of human as well as non-human actors which the residents have accounted for.

The second challenge moves the issue from the struggle between persuasion and negotiation to the dichotomy between persuasion and manipulation. This is an ethical question because, although both strategies seek influence on attitudes and behaviours, persuasion is understood as something positive while manipulation is viewed as negative or unfair. Consequently, technologies based on persuasive design methods can be considered ethically sound while manipulative design is wrong. The matter is further complicated as it is difficult to fully inform about the consequences of being surveilled, because surveillance is practiced in many different arenas (public/private) and situations (hanging out with friends, being at a party, at school, at home, at work and so forth).

To avoid manipulation, it is necessary to carefully evaluate the approaches motivating the design process as well as the developed methods of persuasion as part of a network with a diversity of other agencies. Here we suggest that the principle guiding such an evaluation should be the identification of appropriate and inappropriate approaches in the network and influence on a case-by-case basis. Moreover, we argue that it is necessary to identify the residents' negotiations of the surveillance technology and the diverse agencies and actors they account for. Using the concept of participatory surveillance, we argue that an

analysis of the residents' negotiations needs to consider the types of reconfigurations at play in order to decide if and how the residents are empowered.

4.2. User-generated Maps as Participatory Surveillance

Both the persuasion/negotiation struggle and persuasion/manipulation dichotomy is, of course, related to issues of surveillance. The mappings of marginalised residential areas are surveillance practices, as it is necessary to monitor the participating residents to create maps of their activities and experiences in the local area. Consequently, the discussions of negotiation and manipulation in relation to persuasion allow us to understand our surveillance practices in a way that, hopefully, empower residents and thus provide an ethically sound mapping process – what we call networked persuasion.

To more adequately understand the empowering potentials of the mappings in marginalised residential areas, we suggest the concept of participatory surveillance. This is not to belittle the inherent issues of privacy invasion and repressive control. Such issues are serious challenges for the actors involved in designing the technologies, especially when the explicit aim is to influence and change attitudes and behaviour as in this case. However, as we discussed in the above, participatory surveillance is a particular perspective that offers the possibility of bringing to light other aspects of the surveillance practices than, for example, a privacy- or "Big Brother" perspective might provide. Thus, our point is certainly not that inappropriate or even dangerous potentials should be ignored, but instead that participatory surveillance is a way to broaden our understanding of maps as surveillance technologies.

The concept of participatory surveillance makes it possible to interpret the struggle between persuasion and negotiation as a potentially positive, constructive process. This power struggle is also about appropriating the technologies, i.e. it is a way for the residents to make the maps "their own" by reconfiguring and "hacking" the technologies in different ways, which is not necessarily predictable for the different actors, including the designers. Such productive negotiations and reconfigurations provide the basis for power relations that empower users, as the maps are domesticated and customised by the residents themselves.

Consequently, the participatory surveillance perspective on mappings of marginalised residential areas can be helpful with regards to the dichotomy between persuasion and manipulation. The guiding principle for determining if the designed technologies are persuasive or manipulative could be based on how well the residents are empowered. Similarly, the technology design becomes manipulative if maps cannot be reconfigured in a constructive way, but instead leads to either passive or aggressive resistance. Accordingly, we suggest an iterative process of evaluating the residents' negotiations of the mapping technology. If the negotiation is, for example, violent or generally unproductive it is likely that the residents feel that they are being forced or manipulated rather than supported and motivated. This means that we constantly have to rethink and restructure the mapping in relation to the residents' reconfigurations of the technologies.

Working with the concepts of networked persuasion and participatory surveillance, we have suggested an iterative mapping process of evaluating stakeholders' persuasive approaches as well as the residents' negotiations as part of a network of negotiations. This is done to avoid unethical manipulation as well as to evaluate and include the residents' negotiations in the mapping process. This framework stresses a certain way of designing persuasive technologies where we try to avoid a determinist approach of building the "perfect" map that makes it possible to intervene at the "opportune" moment; the map should not be thought of as just an inert instrument, but as participative surveillance technology, which is potentially empowering for the residents in marginalised residential areas.

4.3. Reclaiming the Neighbourhood

To identify and evaluate the residents' responses to or negotiations of the mapping technologies, we need to develop a mapping method that enables us to monitor people and visualise the results in maps. The visualisation method we suggest has two levels of application. First, it is a theoretical framework (networked persuasion) that can make visible conditions in the persuasive process to create possibilities of monitoring and/or increase group communication (Suchman 1995; Star & Strauss 1999). This is necessary in the process of visualizing negotiations and to investigate how maps can be used as monitoring technologies to involve people in the development of their neighbourhoods. Our theoretical starting point makes it possible to explore how mobile monitoring technologies (e.g. GPS-enabled smartphones) are negotiated when they are introduced into a new area. Moreover, it is interesting to study how the monitoring situation changes when the person carrying the technology moves from one place to another. Thus, the concept of networked persuasion is suitable to analyse how the use of mapping technologies can weaken as well as strengthen different types of engagement with the surroundings.

Second, the visualisation method is a concrete GPS-software for smartphones that enables us to make maps of people's locations and experiences. As mentioned earlier, we have chosen to map the residents' locations and experiences. However, this does not mean that we exclude other actors from our accounts. In our perspective, the maps are not the boundaries of our analysis. Rather, they are part of a network of other actors, such as the police, the landlords, the municipality, and the buildings. Even though these actors are not the ones wearing and using the surveillance technology we will still track them analytically if they act in relation to the marginalised residential area. The residents are asked to carry smartphones that enables them to register and share locations, activities and experiences (Fullpower Technologies 2010; FitnessKeeper 2010; MapMyFitness 2009; Schneider 2010; Tinderhouse Limited 2010). After registering and sharing using smartphones we will invite the residents to workshops, where they are introduced to the maps that they have co-created by carrying GPS-enabled smartphones and sharing information about their experiences and behaviour in the areas they spend time in. Here, it is possible for the residents to create new content for the maps created before the workshops as well as elaborate on their comments to specific locations. The idea is that the mappings can facilitate conversations about the marginalised residential area. By discussing behaviour and experiences in relation to locations, the residents will be able to talk about different actors (e.g. the built environment) in their neighbourhood in a more concrete manner as well as coming up with suggestions for change. This means that the aim is not necessarily to discuss externalities (such as immigration or high levels of unemployment) but instead to allow the residents to be part of a process where they shape their local area in a very concrete manner, namely by creating it on a map. We believe that such maps have the potential to be empowering since the shaping, and subsequently the ownership of, the area are made visible for other stakeholders in the area, such as the municipality, police and landlords. Since smartphones are to be used, a criterion for participating in the tracking is that individuals own this technology. We are aware that this excludes some groups from the mappings. However, we have found that smartphones as well as software that allow tracking and registering experiences and behaviour are widely accessible to the citizens (Sand 2009a, 2009b, 2009c) and thus that we will be able to find participants for the mappings in these areas.

We focus on negotiations of technologies and this gives us the opportunity to explore participatory and user-involving elements in relation to mapping. As argued earlier, the participatory aspect of surveillance can be, but is not necessarily, empowering, since it can be a way to build and share identities and thus reclaim "copyright" over one's own life (Koskela 2004). This means that we need to develop tools and methods that enable the residents to participate in the surveillance in a constructive way, one which still engages with their personal projects and allows them to share their experiences from the neighbourhood. Here, we are inspired by theories in the field of user-involving design, which enables the planning and carrying out of a design process that combines aspects of democracy and creativity (Kanstrup and Christiansen 2006, 2009), and puts focus on the design process as a social activity that is as important as the end product – in this case, a map (Brandt 2006). Consequently, both the surveillance activities and the

creation of the maps are part of the user-involving process. This means that access to negotiations that take place in the design process can be gained by participatory observation and interviews when the residents are under surveillance and during the workshops.

In the mappings of marginalised residential areas we work with a small-scale design process. We focus on how to continuously trigger smaller changes towards a specific goal rather than planning for one big intervention that fits everyone at all times. In relation to this, the maps serve as platforms from which it is possible for the residents to negotiate changes and thus participate in the process of developing the areas.

Using the developed framework, the process of mapping marginalised residential areas can be designed in ways that facilitate constructive ways of negotiating surveillance, by welcoming residents to reconfigure the maps into technologies for constructing and sharing identities. Consequently, mappings in marginalised residential areas can work as a potentially empowering platform for the residents by inviting them not only to *feel* ownership towards their neighbourhood but also to *reclaim* it by commenting on it in a map. The act of reclaiming ownership of the neighbourhood using these maps changes the role of the residents, as they are not only seen as belonging to the area but are visible in the process of co-shaping it; as a result, they are displayed as responsible for taking care of it. Making this shaping process visible to all actors in the process will hopefully further encourage the empowered residents to take ownership of their neighbourhood.

5. Conclusion

In this article, we have explored ways to empower residents in marginalised areas. Networked persuasion is suggested as a concept through which to evaluate persuasive approaches and thus avoid manipulation. Moreover, negotiation is discussed as a creative input to the design process, and we suggest participatory surveillance as a perspective from which positive and constructive aspects of the monitoring practices in marginalised residential areas can be emphasised. Our point is not to exonerate persuasive technologies, in particular mapping technologies, from ethical problems. Rather, we offer a perspective that adds to our understanding of design processes that seek to facilitate persuasive interventions that empower residents.

We have developed a theoretical framework that allows us to proceed with the empirical set-up, which we have discussed in this article. This preliminary work makes it possible for us to facilitate a development process that include the residents' negotiations and reconfigurations of the technologies in marginalised areas rather than force upon them a "perfect", pre-designed plan. Additionally, our framework and eventually our empirical findings can, hopefully, be of value to other urban development projects as well as provide inspiration for user-driven design projects in general.

Lastly, we want to point to some limitations in our theoretical framework and empirical set-up. The set-up we have suggested is useful in creating maps that show snapshots of situations. In this way, the maps make it possible to enter action and see how the residents are involved in and understand their neighbourhood. However, the involvement and understanding of the local area is shaped to a far extent by the negotiations taking place between different persons and not just the individual. These negotiations are difficult to access from these maps. Moreover, the momentary pictures provided by the comments to certain locations make it difficult to trace agencies over time. This means that we cannot rely solely on the maps to trace negotiations but also have to use the other methods we have mentioned: workshops, qualitative interviews and participative observation. What this means is that, rather than being tools to show everything, the maps become tools for negotiation and communication. Here, the creation of the map can be used as a means to facilitate negotiations as well as collaboration between the members in the group in relation to how they are involved in and understand the marginalised residential area. Exactly the collaborative and social aspect of creating maps might in itself facilitate a persuasive process towards involving people in taking care of the area they live in.

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